CHAPTER 3

CLASSIFICATION OF ELEMENTS AND PERIODICITY IN PROPERTIES

CHAPTER ESSENCE AT A GLANCE:
Unifies periodic table organization through atomic numbers and explains periodic variations in radii, ionization energy, and electronegativity.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

- **Modern Periodic Law:** Physical and chemical properties of elements are periodic functions of their atomic numbers.
- **Atomic & Ionic Radii Trends:** Size decreases across periods due to increasing effective nuclear charge and increases down groups with added shells.
- **Ionization Enthalpy:** Energy needed to remove the outermost valence electron; increases across periods and decreases down groups.
- **Electronegativity:** Relative electron-attracting power in a chemical bond, peaking at fluorine in the top-right of the periodic table.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$Z_{\text{eff}} = Z - \sigma$$

Effective Nuclear Charge and Slater Shielding

$$\Delta_i H_1 < \Delta_i H_2 < \Delta_i H_3$$

Successive Ionization Enthalpies

$$r_{\text{cation}} < r_{\text{atom}} < r_{\text{anion}}$$

Relative Radii of Ions and Parent Atom

$$\text{IE } ¤ext\& \text{ EN increase } \rightarrow, \quad \text{Atomic radius increases } \downarrow$$

General Periodic Table Trends

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

- **Alert:** Nitrogen has higher first ionization enthalpy than Oxygen due to its stable half-filled 2p3 configuration.
- **Alert:** Chlorine has more negative electron gain enthalpy than Fluorine because of strong inter-electronic repulsion in compact 2p subshell of Fluorine.
- **Alert:** Noble gases have the highest positive electron gain enthalpies because the added electron must enter the next principal energy level.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. State the Modern Periodic Law. How does it overcome the major anomaly of Mendeleev's periodic system regarding atomic masses? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Modern Periodic Law:** The physical and chemical properties of elements are periodic functions of their atomic numbers (nuclear charge Z). **[1 Mark]**

• **Overcoming Anomalies:** Mendeleev organized elements by atomic masses which failed to explain inverted pairs (Ar before K, Co before Ni) and isotopes. Modern periodic law places elements by atomic number, resolving these discrepancies naturally. **[1 Mark]**

• **Key Exam Result:** Atomic number is the fundamental characteristic determining periodic recurring properties.

2. Why is the first ionization enthalpy of Nitrogen higher than that of Oxygen, despite Oxygen having higher nuclear charge? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Electronic Configurations:** Nitrogen (Z=7) is 1s2 2s2 2p3; Oxygen (Z=8) is 1s2 2s2 2p4. **[1 Mark]**

• **Subshell Stability Reasoning:** Nitrogen has an exactly half-filled 2p subshell (2px1 2py1 2pz1) with extra exchange stability. Removing an electron from oxygen gives a stable half-filled 2p3 state and relieves electron repulsion in 2px2, making ionization easier for oxygen. **[1 Mark]**

• **Key Exam Result:** Extra stability of half-filled 2p3 subshell gives Nitrogen higher first ionization enthalpy.

3. Why does Chlorine have a more negative electron gain enthalpy (-349 kJ/mol) than Fluorine (-328 kJ/mol)? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Compact 2p Size of Fluorine:** Fluorine has an extremely small 2p subshell where electron density is exceptionally high. **[1 Mark]**

• **Inter-electronic Repulsion:** An incoming electron enters the compact 2p subshell of fluorine experiencing severe inter-electronic repulsion, lowering energy released compared to the larger, roomier 3p subshell of chlorine. **[1 Mark]**

• **Key Exam Result:** Chlorine possesses the most negative electron gain enthalpy in the entire periodic table.

4. What are isoelectronic species? Arrange the following ions in increasing order of ionic radii and explain your reasoning: N^{3-}, O^{2-}, F^-, Na^+, Mg^{2+}, Al^{3+}. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Definition:** Chemical species (atoms or ions) containing the identical number of electrons are called isoelectronic species (all 6 species have 10 electrons). **[1 Mark]**

• **Nuclear Charge Effect:** Across isoelectronic species, as positive nuclear charge (Z) increases, valence electrons are pulled closer to the nucleus, decreasing ionic radius. **[1 Mark]**

• **Correct Order & Valuation Step:** Al^{3+} (Z=13) < Mg^{2+} (Z=12) < Na^+ (Z=11) < F^- (Z=9) < O^{2-} (Z=8) < N^{3-} (Z=7). **[1 Mark]**

• **Key Exam Result:** Ionic radius decreases with increasing nuclear charge for isoelectronic ions.

5. Define electronegativity. Contrast it with electron gain enthalpy, and explain its trend across Period 3 of the periodic table. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Definition of Electronegativity:** The qualitative measure of the ability of an atom in a bonded chemical molecule to attract shared electron pairs towards itself. **[1 Mark]**

• **Contrast with Electron Gain Enthalpy:** Electronegativity applies to bonded atoms in molecules (dimensionless relative scale); electron gain enthalpy is an absolute energetic quantity measured for isolated gaseous atoms. **[1 Mark]**

• **Period 3 Trend:** Electronegativity increases steadily from Na (0.9) to Cl (3.0) across Period 3 due to decreasing atomic size and increasing effective nuclear charge. **[1 Mark]**

• **Key Exam Result:** Fluorine has the highest electronegativity (4.0 on Pauling scale) in the periodic table.

6. What is meant by diagonal relationship in the periodic table? Give two chemical similarities between Lithium and Magnesium. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Definition:** The remarkable similarity in physical and chemical properties between certain lighter elements of the second period and elements situated diagonally below them in the third period (Li-Mg, Be-Al, B-Si). **[1 Mark]**

• **Underlying Cause:** Arises due to comparable polarizing power (charge divided by radius ratio) and similar electronegativity values. **[1 Mark]**

• **Similarities:** (1) Both Li and Mg form simple normal monoxides (Li_2O and MgO) when burned in oxygen; (2) Chlorides of both (LiCl and $MgCl_2$) are deliquescent and soluble in organic solvents. **[1 Mark]**

• **Key Exam Result:** Li and Mg display diagonal relationship because of almost identical ionic potential.